have entries for several centroids, and each centroid will have points assigned to it across several mappers: that's why in this MR job we also need one reducer per canopy (i.e. per centroid).

Listing 13.4 MapReduce canopy clustering

```
function MRcanopyClustering(points, T1, T2, numShards)
                  #1
  shards ← randomShard(points,
numShards)                                #2
  mappers ← initMappers(numShards, canopyClustering,
shards)          #3
  reducers ← initReducers(1,
canopyClustering)                                    #4
  centroids ← mapReduce(T1, T2, mappers, reducers)
                   #5

mappers.setMethod(classifyPoints)
               #6
  reducers ← initReducers(|centroids|,
join)                                     #7
  canopies ← mapReduce(T1+T2, mappers,
reducers)                                 #8
  return (canopies, centroids)
                                              #9
```

#1 Method `MRcanopyCentroids` takes a list of points, the number of desired centroids, and two thresholds, `T1` and `T2`, to be used for canopy clustering initialization step. It also takes the number of shards

#2 Break down the dataset into `numShards` random shards. This is usually done automatically by the master node of